Circles & Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The equation of a circle**

Write the equation of a circle with center $(-4, 7)$ and radius 9.

$(x+4)^2 + (y-7)^2 = 81$.

2 **[Foundational]** • **Target: The equation of a circle**

The circle below has the radius shown. Find its area and circumference, in terms of π.

Area $= \pi(6)^2 = 36\pi$. Circumference $= 2\pi(6) = 12\pi$.

3 **[Foundational]** • **Target: The equation of a circle**

A circle has diameter 26 and is centered at the origin. Write its equation, and state its radius.

Radius $= \frac{26}{2} = 13$. Equation: $x^2 + y^2 = 169$.

4 **[Foundational]** • **Target: Completing the square**

Find the center and radius of $x^2 + y^2 - 6x + 8y - 11 = 0$.

$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16 = 36 \Rightarrow (x-3)^2 + (y+4)^2 = 36$. Center $(3, -4)$, radius 6.

5 **[Foundational]** • **Target: Distance, midpoint, and slope**

Find the distance between $(-3, 5)$ and $(4, -7)$, to the nearest hundredth.

$d = \sqrt{(4-(-3))^2 + (-7-5)^2} = \sqrt{49 + 144} = \sqrt{193} \approx 13.89$.

6 **[Foundational]** • **Target: Distance, midpoint, and slope**

Find the midpoint and slope of the segment joining $(2, -6)$ and $(-8, 10)$.

Midpoint $= \left(\frac{2-8}{2}, \frac{-6+10}{2}\right) = (-3, 2)$. Slope $= \frac{10-(-6)}{-8-2} = \frac{16}{-10} = -\frac{8}{5}$.

7 **[Foundational]** • **Target: Distance, midpoint, and slope**

Find the slope of the line through $(-2, 7)$ and $(6, -5)$, and state whether the line rises or falls from left to right.

Slope $= \frac{-5-7}{6-(-2)} = \frac{-12}{8} = -\frac{3}{2}$. Since the slope is negative, the line falls from left to right.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Coordinate proofs

Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle, using slopes.

Slope of $AB = \frac{0-0}{4-0} = 0$ (horizontal). Slope of BC is undefined (vertical, since $x = 4$ for both points). A horizontal segment and a vertical segment are perpendicular, so $\angle B = 90^\circ$, confirming a right triangle.

9 [Intermediate] • Target: Coordinate proofs

Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$, $D(0, 5)$ form a parallelogram, by showing both pairs of opposite sides are parallel.

Slope $AB = \frac{3-2}{4-1} = \frac{1}{3}$. Slope $DC = \frac{6-5}{3-0} = \frac{1}{3}$, so $AB \parallel DC$. Slope $AD = \frac{5-2}{0-1} = -3$. Slope $BC = \frac{6-3}{3-4} = -3$, so $AD \parallel BC$. Both pairs of opposite sides are parallel, so $ABCD$ is a parallelogram.

10 [Intermediate] • Target: Tangent lines to circles

A circle has center $O(2, 3)$ and radius 5. Point $P(2, 12)$ lies outside the circle. Find the length of the tangent from P to the circle, to the nearest hundredth.

$OP = \sqrt{(2-2)^2 + (12-3)^2} = \sqrt{81} = 9$. Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{81 - 25} = \sqrt{56} = 2\sqrt{14} \approx 7.48$.

11 [Intermediate] • Target: Tangent lines to circles

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.

The radius to $(3, 4)$ has slope $\frac{4}{3}$, so the tangent (perpendicular to the radius) has slope $-\frac{3}{4}$. Line: $y - 4 = -\frac{3}{4}(x - 3)$, which simplifies to $3x + 4y = 25$.

12 [Intermediate] • Target: Tangent lines to circles

A circle has center $(0, 0)$ and radius 10. Determine whether the point $(6, 9)$ lies inside, on, or outside the circle.

Distance from center $= \sqrt{6^2 + 9^2} = \sqrt{117} \approx 10.82$, which is greater than the radius 10. The point lies outside the circle.

13 [Intermediate] • Target: Circle theorems

A central angle of 84° subtends an arc. Find the measure of the inscribed angle subtending the same arc.

Inscribed angle $= \frac{1}{2}(84^\circ) = 42^\circ$.

14 [Intermediate] • Target: Circle theorems

A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.

Inscribed angle $= \frac{1}{2}(100^\circ) = 50^\circ$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Arc length and sector area** • **Cross-Lesson: 7.01 Solid Geometry & Similarity**

A circle has radius 12 cm. Find the arc length subtended by a central angle of $\frac{5\pi}{6}$ radians.

$s = r\theta = 12\left(\frac{5\pi}{6}\right) = 10\pi$ cm.

16 **[Challenging]** • **Target: Arc length and sector area** • **Cross-Lesson: 8.01 Statistics & Probability**

The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of π.

$72^\circ = \frac{2\pi}{5}$ radians. Sector area $= \frac{1}{2}r^2\theta = \frac{1}{2}(100)\left(\frac{2\pi}{5}\right) = 20\pi$.

17 **[Challenging]** • **Target: Arc length and sector area** • **Cross-Lesson: 1.01 Advanced Algebra**

A sector of a circle with radius 9 has area 27π. Find the central angle, in radians.

$27\pi = \frac{1}{2}(81)\theta = 40.5\theta \Rightarrow \theta = \frac{27\pi}{40.5} = \frac{2\pi}{3}$.

18 **[Challenging]** • **Target: Conics: ellipses and hyperbolas** • **Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic**

Identify the type of conic and state its center for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$.

An ellipse, centered at $(2, -1)$, with $a^2 = 16$ ($a = 4$, horizontal major axis) and $b^2 = 9$ ($b = 3$).

19 **[Challenging]** • **Target: Conics: ellipses and hyperbolas** • **Cross-Lesson: 3.01 Sequences, Series & Complex Numbers**

Identify the type of conic, its center, and its asymptote slopes for $\frac{(x+3)^2}{4} - \frac{y^2}{9} = 1$.

A hyperbola, centered at $(-3, 0)$, with $a = 2$, $b = 3$. Asymptote slopes: $\pm\frac{b}{a} = \pm\frac{3}{2}$.

20 **[Challenging]** • **Target: A well-designed word problem: a Ferris wheel** • **Cross-Lesson: 4.01 Lines, Triangles & Polygons**

This question has three parts. A circular Ferris wheel has a diameter of 40 m, with its center 22 m above the ground. Set up a coordinate system with the origin at ground level, directly below the center. (a) Write the equation of the circle traced by a passenger cabin. (b) A support cable runs from a ground anchor at $(26, 0)$ to a point of tangency on the wheel. Find the length of this cable, to the nearest hundredth of a meter. (c) Find the height above the ground of a cabin when it is 12 m horizontally from the vertical line through the center.

(a) Center $(0, 22)$, radius 20: $x^2 + (y-22)^2 = 400$. (b) $OP = \sqrt{26^2 + 22^2} = \sqrt{1160} \approx 34.06$ m. Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{1160 - 400} = \sqrt{760} \approx 27.57$ m. (c) With $x = 12$: $12^2 + (y-22)^2 = 400 \Rightarrow (y-22)^2 = 256 \Rightarrow y = 22 \pm 16$, so $y = 38$ m or $y = 6$ m.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The equation of a circle

A circle has center $O(8, 4)$ and radius 20. Point $P(8, 33)$ lies outside the circle. Find the length of the tangent from P to the circle.

a. 29

b. 23

c. 21 ✓

d. 19

$OP = \sqrt{(8-8)^2 + (33-4)^2} = \sqrt{841} = 29$. Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{841 - 400} = \sqrt{441} = 21$.

22 [Foundational] • Target: Completing the square

A circle has radius 8. Find the area of the sector with central angle 60°, in terms of π.

a. $\frac{29}{3}\pi$

b. $\frac{32}{3}\pi$ ✓

c. $\frac{64}{3}\pi$

d. $\frac{35}{3}\pi$

Sector area $= \frac{60}{360} \times \pi(8)^2 = \frac{60}{360} \times 64\pi = \frac{32}{3}\pi$.

23 [Foundational] • Target: Distance, midpoint, and slope

A circle has center $(4, -2)$ and passes through $(13, 10)$. Find the slope of the tangent line to the circle at $(13, 10)$.

a. 5/4

b. 1/4

c. **-3/4** ✓

d. -11/4

Radius slope $= \frac{10+2}{13-4} = 4/3$. Tangent $\perp$ radius, so tangent slope $= -\frac{1}{4/3} = -3/4$.

24 [Foundational] • Target: Coordinate proofs

What is the equation of the circle with centre $(-2, 3)$ and radius 4?

a. $(x+2)^2 + (y-3)^2 = 16$ ✓

b. $(x-2)^2 + (y+3)^2 = 16$

c. $(x+2)^2 + (y-3)^2 = 4$

d. $(x-2)^2 + (y-3)^2 = 16$

$(x-h)^2 + (y-k)^2 = r^2 \Rightarrow (x+2)^2 + (y-3)^2 = 16$.

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25 [Foundational] • Target: Tangent lines to circles

A circle has center $(6, 6)$ and radius 2. Find the shortest distance from the point $(16, 30)$ to the circle.

a. 6

b. 38

c. **24** ✓

d. 29

Distance from point to center $= \sqrt{10^2 + 24^2} = 26$. Since $26 > 2$, the point is outside; shortest distance to the circle $= 26 - 2 = 24$.

26 [Foundational] • Target: Circle theorems

Find the slope of the line through $(-4, -2)$ and $(-7, -4)$.

a. **2/3** ✓

b. -4/3

c. 8/3

d. 13/6

$m = \frac{-4+2}{-7+4} = 2/3$.

27 [Foundational] • Target: Arc length and sector area

The point $(17, y)$ lies on the circle $(x + 3)^2 + (x - 5)^2 = 841$ with $y > 5$. Find y.

a. **26** ✓

b. 33

c. 18

d. 25

$(17 + 3)^2 + (y - 5)^2 = 841 \Rightarrow 20^2 + (y - 5)^2 = 841 \Rightarrow y - 5 = \pm 21 \Rightarrow y = 26$ (taking $y > 5$).

28 [Foundational] • Target: Conics: ellipses and hyperbolas

Line ℓ_1 has slope -2. What is the slope of a line perpendicular to ℓ_1?

a. **1/2** ✓

b. 5/2

c. 1/4

d. 1/6

Perpendicular slopes are negative reciprocals: $m = -\frac{1}{-2} = 1/2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: a Ferris wheel

A circle is given by $x^2 + y^2 - 2x - 16y + 61 = 0$. If its center is (h, k) and radius r, find $h + k + r$.

a. **11** ✓

b. 5

c. 12

d. 7

Complete the square: center $(1, 8)$, $r = 2$. $h + k + r = 1 + 8 + 2 = 11$.

30 [Intermediate] • Target: The equation of a circle

Find the midpoint of the segment joining $(6, -4)$ and $(8, -10)$.

a. (7,-16)

b. (7,-10)

c. (7,-3)

d. **(7,-7)** ✓

Midpoint $= \left(\frac{6+8}{2}, \frac{-4-10}{2}\right) = (7, -7)$.

31 [Intermediate] • Target: Completing the square

Find the distance between $(3, 0)$ and $(-2, -12)$.

a. 11

b. 10

c. **13** ✓

d. 20

$d = \sqrt{(-2-3)^2 + (-12-0)^2} = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

32 [Intermediate] • Target: Distance, midpoint, and slope

Find the radius of the circle $x^2 + y^2 + 4x + 2y - 11 = 0$.

a. **4** ✓

b. 1

c. 9

d. 7

Complete the square: $(x + 2)^2 + (y + 1)^2 = 16$, so $r = 4$.

33 [Intermediate] • Target: Coordinate proofs

A circle has center $(-2, 6)$ and passes through $(18, 27)$. Find its radius.

a. 19

b. 17

c. **29** ✓

d. 24

$r = \sqrt{(18+2)^2 + (27-6)^2} = \sqrt{20^2 + 21^2} = 29$.

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34 [Intermediate] • Target: Tangent lines to circles

Triangle has vertices $A(0, 0)$, $B(10, 0)$, $C(0, 24)$. Find the length of the hypotenuse BC.

a. 37

b. 11

c. **26** ✓

d. 16

$BC = \sqrt{(10-0)^2 + (0-24)^2} = \sqrt{10^2 + 24^2} = 26.$

35 [Intermediate] • Target: Circle theorems

Line ℓ_1 has slope -1. What is the slope of a line parallel to ℓ_1?

a. -7

b. 2

c. **-1** ✓

d. -4

Parallel lines have equal slopes: $m = -1$.

36 [Intermediate] • Target: Arc length and sector area

Find the surface area of a rectangular prism with length 12, width 5, and height 6.

a. 166

b. **324** ✓

c. 131

d. 329

$SA = 2(lw + lh + wh) = 2((12)(5) + (12)(6) + (5)(6)) = 324.$

37 [Intermediate] • Target: Conics: ellipses and hyperbolas

In the diagram below, lines DE and BC are parallel. If AD = 6, DB = 4, and AE = 9, what is the length of EC?

a. **6** ✓

b. 7

c. 8

d. 10

Since DE is parallel to BC, triangle ADE is similar to triangle ABC. By the properties of similar triangles, the ratio of corresponding sides is equal. Therefore, we have the proportion: $\frac{AD}{AB} = \frac{AE}{AC}$. We are given AD = 6, DB = 4, and AE = 9. So, AB = AD + DB = 6 + 4 = 10. Substituting these values into the proportion: $\frac{6}{10} = \frac{9}{AC}$. To find AC, we can cross-multiply: $6 \times AC = 10 \times 9$ $6 \times AC = 90$ $AC = \frac{90}{6}$ $AC = 15$. We are asked to find the length of EC.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: A well-designed word problem: a Ferris wheel** • **Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic**

In the circle shown below, chords AB and CD intersect at point P. If $PA = 4$, $PB = 6$, and $PC = 3$, what is the length of segment $\overline{PD}$?

a. 5

b. 7

c. **8** ✓

d. 9

By the Intersecting Chords Theorem (Power of a Point Theorem), when two chords AB and CD intersect at an interior point P, the products of their segments are equal: $PA \cdot PB = PC \cdot PD$ Substitute the given values into the equation: $(4)(6) = (3)(PD)$ $24 = 3 \cdot PD$ $PD = \frac{24}{3} = 8$ Common error analysis: - Adding the segment lengths instead of multiplying ($PA + PB = PC + PD \Rightarrow 4 + 6 = 3 + PD \Rightarrow PD = 7$) yields 7, which fails because the segments satisfy a multiplicative re.

39 **[Challenging]** • **Target: The equation of a circle** • **Cross-Lesson: 3.01 Sequences, Series & Complex Numbers**

Find the surface area of a cube with edge length 9.

a. 377

b. **486** ✓

c. 188

d. 49

Applying Solid Geometry & Similarity principles, $SA = 6s^2 = 6(9)^2 = 486$. Therefore, the verified correct option is 486.

40 **[Challenging]** • **Target: Completing the square** • **Cross-Lesson: 4.01 Lines, Triangles & Polygons**

$ABCD$ is a parallelogram with $A(4, -6)$, $B(6, -6)$, $C(2, -6)$. Find D.

a. **(0,-6)** ✓

b. (0,-1)

c. (0,-13)

d. (0,-4)

In parallelogram $ABCD$, diagonals bisect each other: $A + C = B + D \Rightarrow D = A + C - B = (4 + 2 - 6, -6 - 6 + 6) = (0, -6)$.

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41 [Challenging] • Target: Distance, midpoint, and slope • Cross-Lesson: 5.01 Trigonometry

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. 25 ✓

b. 50

c. 100

d. 125

The equation of the circle is $x^2 + y^2 = 25$. Let the point of tangency be $P(x_0, y_0)$. The equation of the tangent line to the circle at $P(x_0, y_0)$ is given by $xx_0 + yy_0 = 25$. To find the x-intercept A, we set $y = 0$: $xx_0 = 25 \Rightarrow x = \frac{25}{x_0}$. So, $A = (\frac{25}{x_0}, 0)$. To find the y-intercept B, we set $x = 0$: $yy_0 = 25 \Rightarrow y = \frac{25}{y_0}$. So, $B = (0, \frac{25}{y_0})$. The area of triangle OAB is.

42 [Challenging] • Target: Coordinate proofs • Cross-Lesson: 7.01 Solid Geometry & Similarity

Find the volume of a rectangular prism with length 3, width 13, and height 5.

a. 226

b. 162

c. 183

d. 195 ✓

Applying Solid Geometry & Similarity principles, $V = lwh = (3)(13)(5) = 195$. Therefore, the verified correct option is 195.

43 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 8.01 Statistics & Probability

Find the total surface area of a cylinder with radius 5 and height 5. (In terms of π.)

a. 57π

b. 45π

c. 100π ✓

d. 121π

$SA = 2\pi r^2 + 2\pi rh = 2\pi(5)^2 + 2\pi(5)(5) = 100\pi$.

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44 [Challenging] • Target: Circle theorems • Cross-Lesson: 1.01 Advanced Algebra

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.

a. 25 ✓

b. 30

c. 35

d. 40

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$. Since point $P(a, 4)$ lies on the circle, we have $a^2 + 4^2 = r^2$, so $a^2 + 16 = r^2$. The radius from the origin to $P(a, 4)$ is perpendicular to the tangent line at P. The slope of the radius is $m_{radius} = \frac{4-0}{a-0} = \frac{4}{a}$. The given tangent line is $3x + 4y = k$, which can be rewritten in slope-intercept form as $y = -\frac{3}{4}x + \frac{k}{4}$. The slope of the tangent line is $m_{tangent} = -\frac{3}{4}$. Since the rad.

45 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In the diagram below, points A, B, C, and D lie on the circle. If AB = 6, BP = 8, CD = 4, and AP = x, where P is the intersection of chords AC and BD, find the value of x.

a. 4 ✓

b. 5

c. 6

d. 8

The problem involves the intersecting chords theorem (also known as the power of a point theorem). When two chords intersect inside a circle, the product of the lengths of the segments of one chord is equal to the product of the lengths of the segments of the other chord. Let the chords be AC and BD, intersecting at point P. The theorem states: PA · PC = PB · PD From the problem statement, we are given: PA = 8 PB = 8 PC = 4 We need to find the length of PD, which is denoted by x. Substitute the given va.

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